

Kaushik Gupta

SENIOR SOFTWARE ENGINEER

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Websites, Portfolios, Profiles

- <https://www.linkedin.com/in/kaush-gupta/>
- <https://github.com/Kaush22>

Professional Summary

Software engineering professional with deep expertise in developing robust, scalable applications. Strong focus on team collaboration, driving projects to successful completion, and adapting to evolving requirements. Proficient in multiple programming languages, frameworks, and tools. Values delivering high-quality results and fostering productive work environment.

Experience - 6.1 yrs

Coforge | Senior Software Engineer

Aug 2024 – Present

- Developed a **Specialized Language Model (SLM)** using **GenAI** and **LangChain** for cancer diagnostics, leveraging **NLP** and **retrieval-augmented generation (RAG)** techniques to enhance clinical decision support with accurate and context-rich outputs.
- Created a **machine learning classifier** with **anomaly detection** for Mission Watch, optimizing aviation alert management through **real-time filtering** and **prioritization**.
- Designed and delivered **scalable, AI-powered** applications by applying **agile methodologies** and modular **GenAI** components for **rapid** and **efficient development cycles**.
- Managed **multiple projects** simultaneously while maintaining **strict deadlines** and **high-quality standards**.

Ignitarium Technology Solutions | Senior Developer

Mar 2022 – Aug 2024

- Led development of a rolling stock inspection system using image processing, Faster R-CNN, and UNet models in TensorFlow, achieving 86.3% average precision through hyperparameter tuning and cross-validation for accurate component detection and defect identification.
- Managed Agile teams across multiple sprints, coordinated peer reviews, and ensured high-quality deliverables through continuous integration and iterative releases, demonstrating strong project management and collaboration skills.
- Engineered a drone-based warehouse monitoring system, applying camera calibration, homography, and perspective transformation to align and decode barcodes, ensuring accurate shelf-level inventory updates.
- Enhanced warehouse efficiency by integrating computer vision pipelines with real-time databases using Python, OpenCV, and image processing techniques, significantly improving data accuracy and system responsiveness.

- Developed and executed strategic project plans aligned with business objectives, ensuring effective resource allocation and on-time delivery through strong planning and stakeholder collaboration.
- Engineered deep learning modules for component detection using **Python**, **OpenCV**, **PyTorch**, **NumPy**, and **Pandas**, delivering robust **image analysis** and **defect detection** capabilities.
- Applied advanced **image processing** techniques (rectification, enhancement, normalization) and implemented **Faster R-CNN** and **UNet** models in **TensorFlow**, achieving high detection accuracy through model tuning.
- Led Agile development across multiple sprints, managing **cross-validation**, **hyperparameter tuning**, and progress tracking, while ensuring timely releases and maintaining high code quality and team coordination.

Education

Indian Institute of Information Technology Design and Manufacturing, Chennai, India

Master of Technology, Electronics and Communication Engineering **May 2019**

- Worked as half time teaching assistance, and got experience on different projects on image processing like Passenger counting system using stereo camera, fault detection in nut bolts, etc.
- GPA: 8.44 CGPA

Gurukula Kangri Vishwavidyalaya, Haridwar, India

Bachelor of Technology, Electronics and Communication Engineering, **May 2017**

- Score 464 in gate exam, worked on projects like virtual keypad and CNC machine.
- GPA: 76.13%

Skills & abilities

- | | |
|--------------------------------------|--------------------------------|
| • Python | • LLMs (Large Language Models) |
| • Specialized Language Models (SLMs) | • Prompt Engineering |
| • Agentic AI | • NLP |
| • Anomaly Detection | • Supervised Learning |
| • Unsupervised Learning | • Computer Vision |
| • OpenCV | • PyTorch |
| • Numpy | • Matplotlib |
| • Pandas | • Scikit-learn |
| • Transformers | • SpaCy |
| • NLTK | • Gensim |
| • Accelerate | • DeepSpeed |
| • AWS Sagemaker | • EC2 |

- IAM
- Docker
- Gitlab
- GitHub

Achievements And Extracurricular Activities

- Coaching Champ, Ignitarium Technology Solutions, 02/01/23, For helping juniors throughout the project.
- Spot Award, Ignitarium Technology Solutions, 07/01/22, For Outstanding Performance.
- Best Performer, TCS Ltd, 04/01/21, For outstanding performance.
- On the Spot Award, TCS Ltd, 11/01/20, For stellar contribution in Power Utility project.
- Coordinator, RoSMA, International Conference on Robotics and Manufacturing, IIITDM Kancheepuram
- Volunteer, LIGHT, IIT Kharagpur initiative, NGO
- Conducted HPE Summer training program, Embedded Systems and Robotics, 06/01/16
- Conducted Session, Embedded and Robotics, GKV Haridwar
- GATE Exam, 464, 2017

Technical Summary

- Proficient in SDLC methodologies and Agile practices, including sprint planning, daily stand-ups, and retrospective meetings.
- Expertise in designing and deploying domain-specific SLMs for healthcare, leveraging transformer-based architectures to process and interpret cancer-related clinical text for diagnostic support.
- Extensive experience in computer vision frameworks: OpenCV, PyTorch, TensorFlow, and Sci-kit, adept at developing and deploying deep learning models for various applications.
- Expertise in data annotation tools: Vott, LabelMe, and LabelImg, proficient in generating high-quality annotated datasets for training machine learning models.
- Skilled in Python libraries: NumPy, Pandas, Gensim, Accelerate, DeepSpeed, Matplotlib, Seaborn, Spacy, and NLTK enabling effective data manipulation, analysis, and visualization of large datasets.

Personal Details

Date of Birth: 20/Dec/1994

Nationality: Indian

Marital Status: Single

Gender: Male